



CSE 4513

Lec – 5

User interface Design



THE \$300 MILLION BUTTON



- ✓ user filled their shopping cart with products and pressed the **Checkout** button
- ✓ a form with two fields (email, pass), two buttons (login, register), and one link(forgot pass) comes before they could actually enter the information to pay for the product.
- ✓ How could they have problems with it?

Even some 1st time shopper said **“I’m Not Here To Be In a Relationship, I just want to buy something.”**

Some 1st time shoppers couldn’t remember if it was their first time, becoming frustrated as each common email and password combination failed.

Many vocalized how the retailer only wanted their information to send them with marketing messages they didn’t want.

Repeat customers weren’t any happier. Except for a very few who remembered their login information, most stuck on the form.

couldn’t remember the email address or password they used. Remembering which email address they registered with was problematic – many had multiple email addresses or had changed them over the years.

THE \$300,000,000 FIX



- ✓ designers fixed the problem simply.
- ✓ They took away the *Register* button
- ✓ In its place, they put a *Continue* button with a simple message:

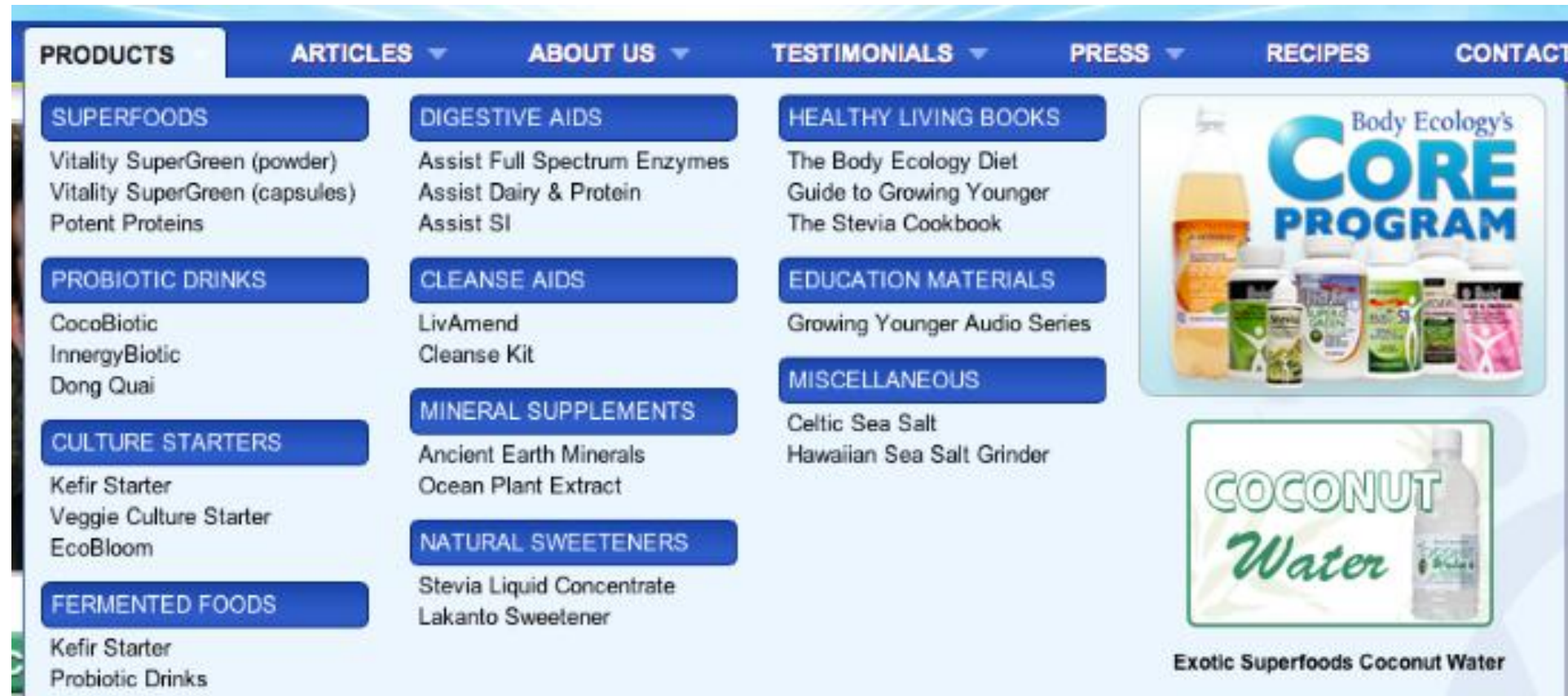


The results: The number of customers purchasing went up by 45%. The extra purchases resulted in an extra \$15 million the first month. For the first year, the site saw an additional \$300,000,000.

REMOVING DROP-DOWN INCREASES REVENUE 56.43%



Body Ecology is an e-commerce store that sells diet products. They employed a drop-down menu to help narrow down the customers' choices

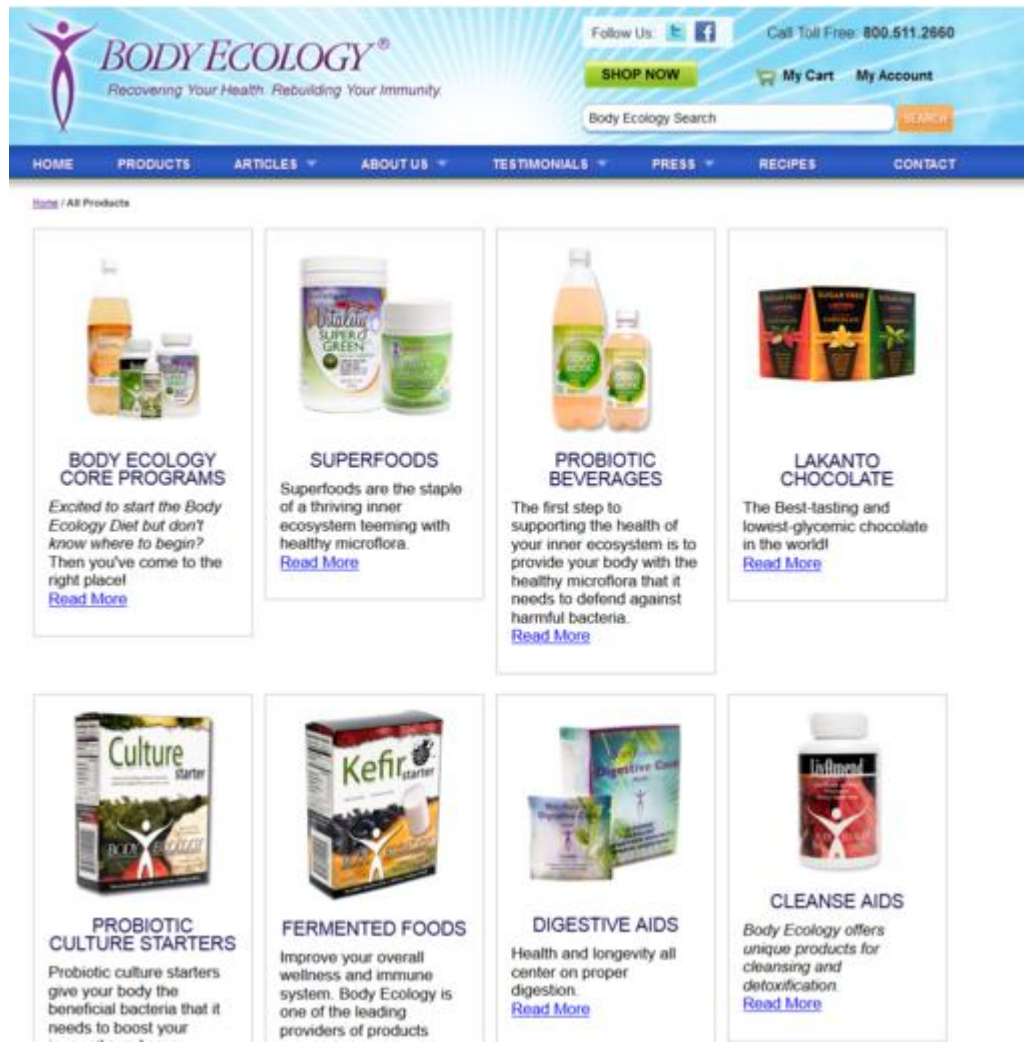


the drop-down menu gives only text links to products. It doesn't display a picture of the product

REMOVING DROP-DOWN INCREASES REVENUE 56.43%



wanted to see if removing this drop-down menu and replacing it with a products page would increase order conversion. This is the replacement:



The product page beat the drop-down menu by **56.43%**, resulting in a total increase in revenue of **\$8,880** over a **two-week** period.

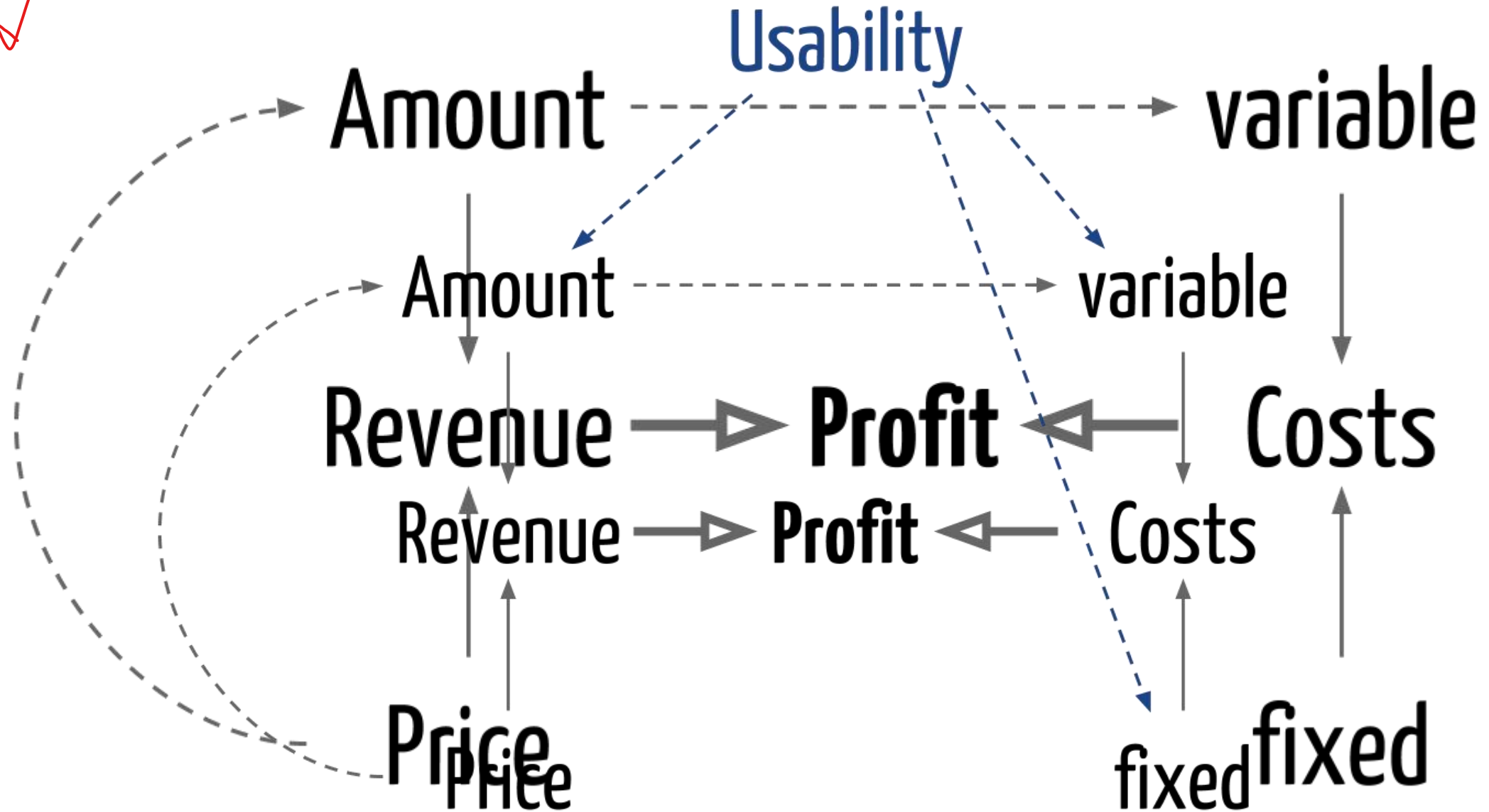
1983

Stay out of the garbage!

You say you have the garbage!

for registering! We appreciate your
your software, you will be sent an e
ave received the key then you will b
nd you can then proceed with the a

HOW USABILITY IMPACTS PROFIT



WHY IS USABILITY IMPORTANT?



Poor usability results in

- ✓ anger and frustration
- ✓ decreased productivity in the workplace
- ✓ higher error rates
- ✓ physical and emotional injury
- ✓ equipment damage
- ✓ loss of customer loyalty
- ✓ costs money

WHAT IS USABILITY?



Usability is a measure of the effectiveness, efficiency and satisfaction with which specified users can achieve specified goals in a particular environment.



THE USER INTERFACE



The user interface is the way for the users to communicate with the computer system and/or the application system.

- ✓ User interfaces should be designed to match the skills, experience and expectations of its anticipated users.
- ✓ System users often judge a system by its interface rather than its functionality.
- ✓ A poorly designed interface can cause a user to make catastrophic errors.
- ✓ Poor user interface design is the reason why so many software systems are never used.

HUMAN FACTORS IN INTERFACE DESIGN



✓ Limited short-term memory

- People can instantaneously remember about 7 items of information. If you present more than this, they are more liable to make mistakes.

✓ People make mistakes

- When people make mistakes and systems go wrong, inappropriate alarms and messages can increase stress and hence the likelihood of more mistakes.

✓ People are different

- People have a wide range of physical capabilities. Designers should not just design for their own capabilities.

✓ People have different interaction preferences

- Some like pictures, some like text.

UI DESIGN: THINGS TO CONSIDER



✓ Assume users

- Have not read the manual
- Have not attended training
- Do not have external help readily at hand

So...

- ✓ All controls should be **clear** and **understandable** and placed in an **intuitive location** on the screen.

UI DESIGN: THINGS TO CONSIDER



- ✓ UI design must take account of the needs, experience and capabilities of the system users.
- ✓ Designers should be aware of people's physical and mental limitations (e.g. limited short-term memory) and should recognise that people make mistakes.
- ✓ UI design principles serve as the foundation for creating interface designs, but it's important to note that not all principles are relevant to every specific design.

USER INTERFACE DESIGN PRINCIPLES



✓ User familiarity

- The interface should be based on **user-oriented terms and concepts** rather than computer concepts.

For example, an office system should use concepts such as letters, documents, folders etc. rather than directories, file identifiers, etc.



✓ Consistency

- The system should **display an appropriate level of consistency**.

Commands and menus should have the same format, command punctuation should be similar, etc.

✓ Minimal surprise

- If a command operates in a known way, the user should be able to **predict the operation of comparable commands**



USER INTERFACE DESIGN PRINCIPLES



✓ Recoverability

- The system should provide some resilience to user errors and allow the user to recover from errors. This might include an undo facility, confirmation of destructive actions, 'soft' deletes, etc.

✓ User guidance

- Some user guidance such as help systems, on-line manuals, etc. should be supplied

✓ User diversity

- Interaction facilities for different types of user should be supported. For example, some users have seeing difficulties and so larger text should be available

DESIGN ISSUES IN UIs



✓ Two problems must be addressed in interactive systems design

➤ How should information from the user be provided to the computer system?

➤ How should information from the computer system be presented to the user?

INTERACTION STYLES



Interaction style	Main advantages	Main disadvantages	Application examples
<u>Direct manipulation</u>	Fast and intuitive interaction Easy to learn	May be hard to implement Only suitable where there is a visual metaphor for tasks and objects	<u>Video games</u> <u>CAD systems</u>
<u>Menu selection</u>	Avoids user error Little typing required	Slow for experienced users Can become complex if many menu options	<u>Most general-purpose systems</u>
<u>Form fill-in</u> Easy to learn Checkable	Simple data entry	Takes up a lot of screen space Causes problems where user options do not match the form fields	<u>Stock control</u> <u>Personal loan processing</u>
<u>Command language</u>	Powerful and flexible	Hard to learn Poor error management	<u>Operating systems</u> <u>Command and control systems</u>
<u>Natural language</u>	Accessible to casual users Easily extended	Requires more typing Natural language understanding systems are unreliable	<u>Information retrieval systems</u>

USER CENTERED DESIGN (UCD)

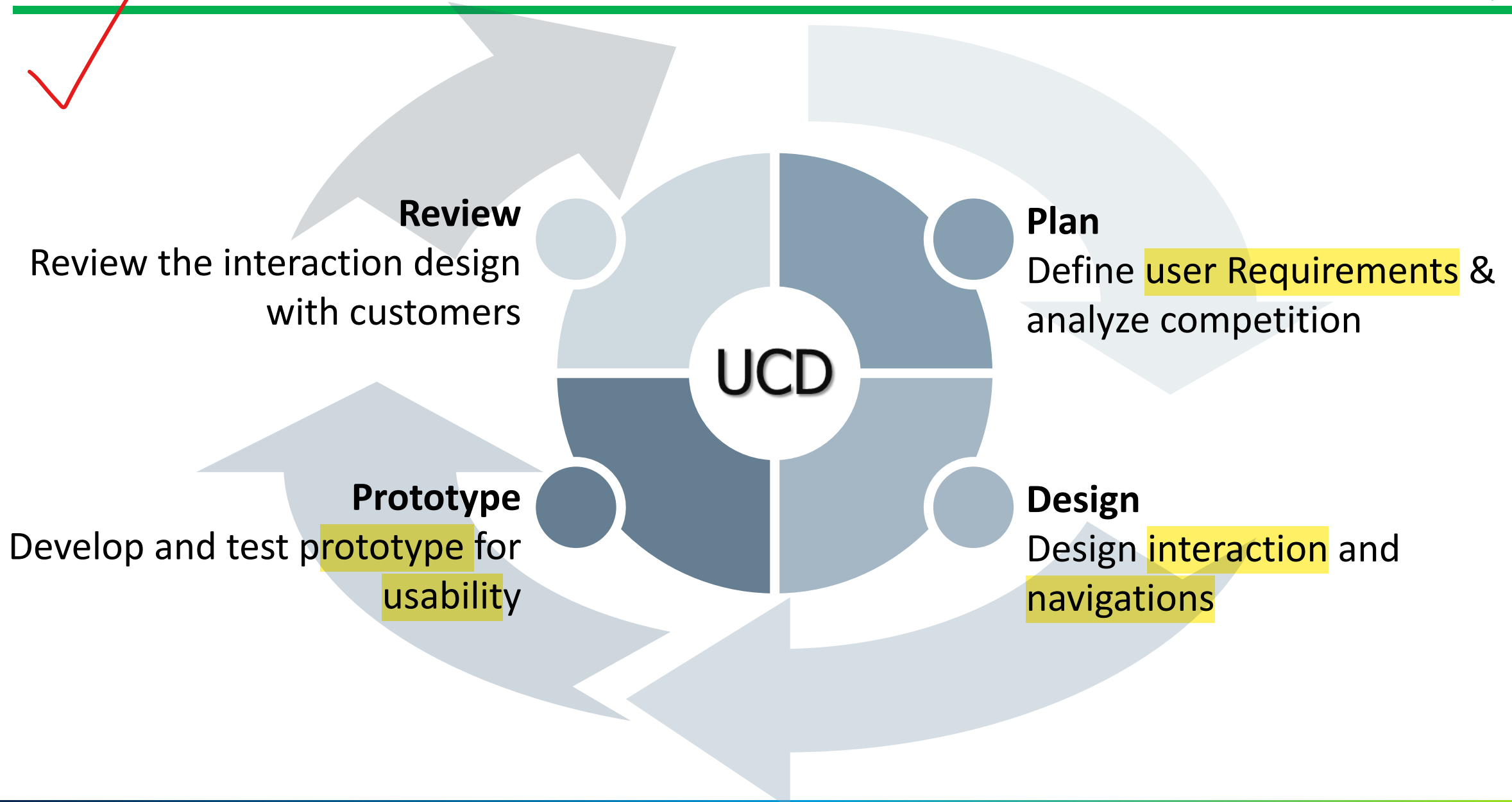


✓ UCD is a dialog between the customer and the designer

✓ Rules of thumb:

- Get to know and understand the users.
- Build an application, applying usability principles.
- Test designs by observing users in a real work setting (environment and work load).

USER CENTERED DESIGN PROCESS



EXAMPLE : PAPER PROTOTYPE



UI DESIGN IS MULTI DISCIPLINARY



✓ A team includes

- Analyst
- Designer
- Technology expert
- Graphic artist
- Social and behavioral scientist
- Programmer

GOLDEN RULES FOR BETTER INTERFACE DESIGN



1. Strive for Consistency

- ✓ Designing “consistent interfaces” means using the same design patterns and the same sequences of actions for similar situations.
- ✓ a consistent interface will allow your users to complete their tasks and goals much more easily.

The screenshot displays the Amazon product page for the book "The New ABCs of Research: Achieving Breakthrough Collaborations Reprint Edition, Kindle Edition" by Ben Shneiderman. The page layout is consistent with standard Amazon product listings, featuring a book cover, title, author information, pricing, and purchase options.

Book Details:

- Title:** The New ABCs of Research: Achieving Breakthrough Collaborations Reprint Edition, Kindle Edition
- Author:** Ben Shneiderman (Author)
- Format:** Kindle Edition
- Ratings:** 16 ratings (5 stars)

Pricing and Purchase Options:

Format	Price (CDN\$)	Used from (CDN\$)	New from (CDN\$)
Kindle Edition	13.71	-	-
Hardcover	45.63	40.52	45.63
Paperback	26.50	22.35	26.50

Additional Information:

- Digital List Price:** ~~CDN\$ 18.99~~
- Print List Price:** ~~CDN\$ 26.50~~
- Kindle Price:** **CDN\$ 13.71**
- Savings:** Save CDN\$ 12.79 (48%)
- Includes:** free international wireless delivery via Amazon Whispernet
- Buttons:** Buy now with 1-Click, Send a free sample, Add to Wish List

Author Information:

- Follow the Author:** Ben Shneiderman
- Follow Button:** + Follow

Product Description:

The problems we face in the 21st century require innovative thinking from all of us. Be it students, academics, business researchers or government policy makers. Hopes for improving our healthcare, food supply, community safety and environmental sustainability depend on the pervasive application of research solutions.

Additional Features:

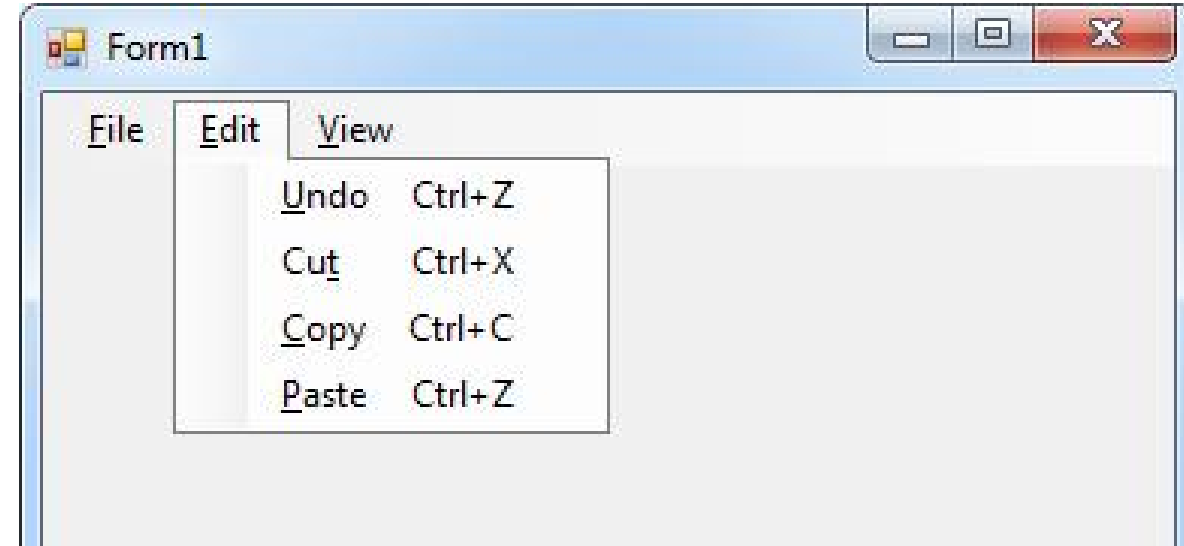
- Length:** 338 pages
- Enhanced Typesetting:** Enabled
- Page Flip:** Enabled
- See all supported devices:** [Dropdown]

GOLDEN RULES FOR BETTER INTERFACE DESIGN



2. Enable Frequent Users to Use Shortcuts

- ✓ Users will benefit from shortcuts if they need to complete the same tasks often.
- ✓ Expert users might find the following features helpful:
 - Abbreviations
 - Function keys
 - Hidden commands
 - Macro facilities

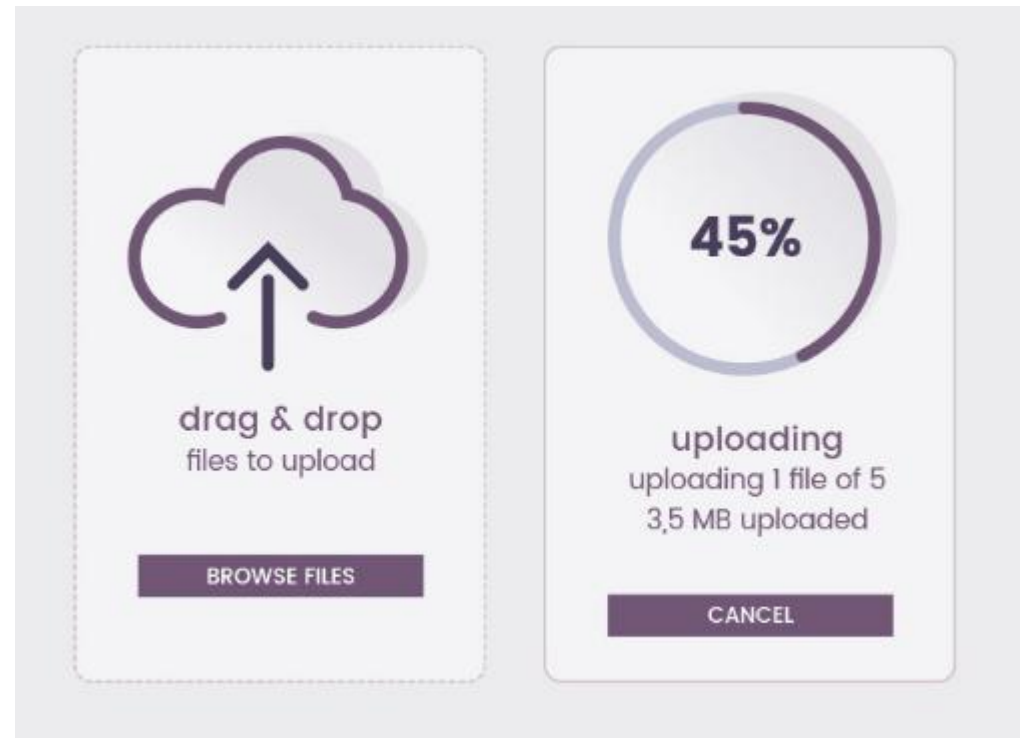


GOLDEN RULES FOR BETTER INTERFACE DESIGN



3. Offer Informative Feedback

- ✓ need to **keep users informed** of what is happening at every stage of their process.
- ✓ This feedback needs to be **meaningful, relevant, clear, and fit the context.**



GOLDEN RULES FOR BETTER INTERFACE DESIGN



4. Design Dialog to Yield Closure

- ✓ Sequences of actions need to have a **beginning, middle and end.**
- ✓ Your user must see the path in his action
- ✓ by offering him the end of an interaction through feedback you **reduce his mental load and improve his experience** on your interface.
- ✓ Don't keep them wondering!

Thank you, your order has been placed.

Please check your email for order confirmation and detailed delivery information or visit [Message Centre](#) to review your notifications.



Tomorrow, March 16
Estimated delivery



GOLDEN RULES FOR BETTER INTERFACE DESIGN



5. Offer Simple Error Handling

- ✓ A good interface should be designed to avoid errors as much as possible.
- ✓ But when errors do happen, your system needs to make it easy for the user to understand the issue and know how to solve it.
- ✓ Simple ways to handle errors include displaying clear error notifications along with descriptive hints to solve the problem.

Personal details

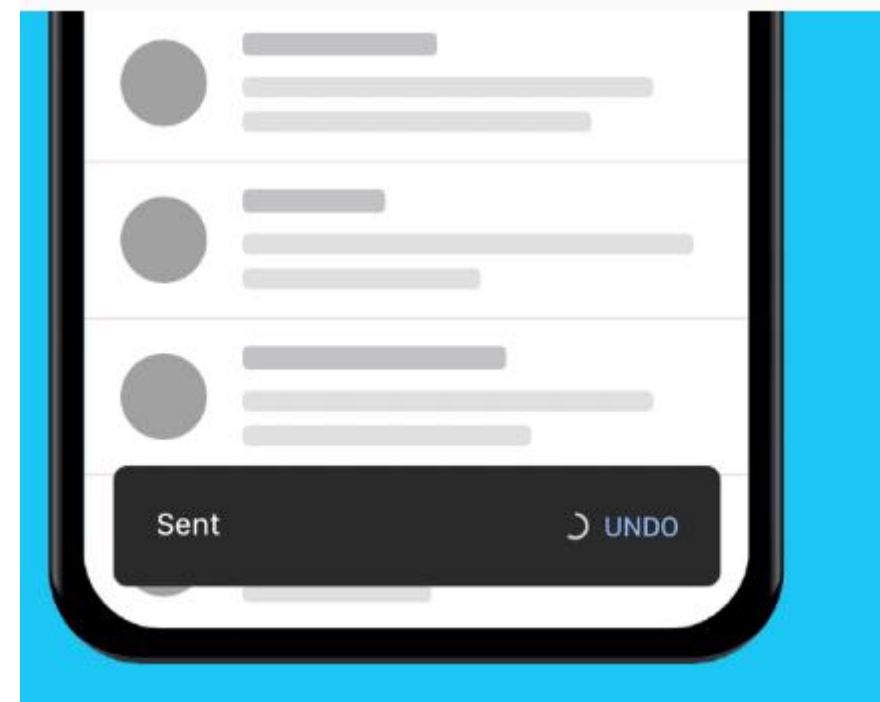
Jason	Last name
user@litmuscom	
Please enter a valid email address	
*****	Confirm password
Great choice!	
Which email service provider do you use?	

GOLDEN RULES FOR BETTER INTERFACE DESIGN



6. Permit Easy Reversal of Actions

- ✓ It's an instant relief to find that “undo” option after a mistake is made.
- ✓ Users will feel less anxious and more likely to explore options if they know there's an easy way to reverse any accidents.
- ✓ This rule can be applied to any action, group of actions, or data entry.



GOLDEN RULES FOR BETTER INTERFACE DESIGN



7. Support Internal Locus of Control

- ✓ It's important to give **control and freedom to your users** so they're able to feel they're in charge of the system, not the other way round.
- ✓ **Users should be the initiators of the actions rather than the responders.**



Tom Burns
3.28K subscribers

HOME

VIDEOS

PLAYLISTS

COMMUNITY

CHANNELS

ABOUT



My other channels

SUBSCRIBED



All



Personalized



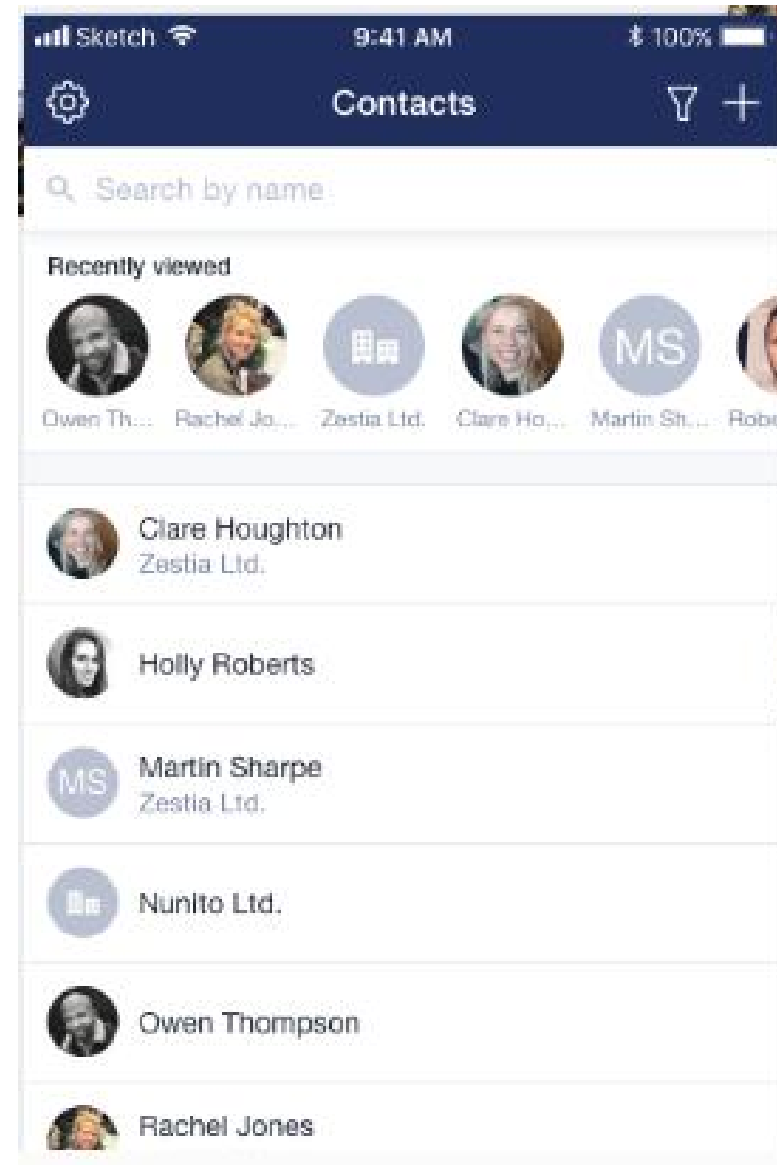
None

GOLDEN RULES FOR BETTER INTERFACE DESIGN



8. Reduce Short-Term Memory Load

- ✓ If we keep our interfaces simple and consistent, obeying to patterns, standards and conventions, we are already contributing to better recognition and ease of use.
- ✓ There are several features we can add to aid our users depending on their goals. For example, in an ecommerce environment, a list of recently viewed or purchased items.
- ✓ don't make your user remember more things than necessary.



EXERCISE



You are using many mobile applications. Surely you are not happy with all User interfaces of the apps you are using. Choose one of such app with poor user interface. And Redesign the User Interface for better usability. Write 5 sentence why your designed UI is more usable that the existing one.

USER INTERFACE EVALUATION



- ✓ Some evaluation of a user interface design should be carried out to assess its suitability.
- ✓ Full scale evaluation is very expensive and impractical for most systems.
- ✓ Ideally, an interface should be evaluated against a usability specification. However, it is rare for such specifications to be produced.

Attribute	Description
<u>Learnability</u>	How long does it take a new user to become productive with the system?
<u>Speed of operation</u>	How well does the system response match the user's work practice?
<u>Robustness</u>	How tolerant is the system of user error?
<u>Recoverability</u>	How good is the system at recovering from user errors?
<u>Adaptability</u>	How closely is the system tied to a single model of work?

SIMPLE EVALUATION TECHNIQUES



✓ Less expensive techniques for user interface evaluation:

- Questionnaires
- Observations of users at work
- Video snapshots of typical uses
- Software components that gather information on the usage of the user interface
- Software components that allow direct feedback from users

“Walk a mile in the user’s shoes”

NEXT CLASS



SOFTWARE DESIGN